

Fault model: $n \geq 3f + s + 1$ (Byzantine f , soft-fault s ; reputation affects selection only)

PHASE 1: PROPOSE

Primary P (round-robin)

$\pi = (\text{task_id}, \text{result},$
 $\text{confidence}, \text{trace})$

PHASE 2: VALIDATE

Re-solve and verify π ;
deterministic: error-free
semantic: m -verifier vote

PHASE 3: COMMIT

$\phi(\pi) = |\text{ACCEPT}|$
 $- 0.5 |\text{REJECT}|$;
ACCEPT iff $\phi \geq \theta_{\text{accept}}$

ACCEPT

PENDING

REJECT

PENDING (2 rounds) or 2Δ timeout $\rightarrow$ View Change ($2f+1$ confirmations, new primary)